JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

EXAMINATION BRANCH & SRI SIVANI COLLEGE OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

★ SEMESTER END QUESTION PAPERS WITH SOLVED BLUEPRINT ★

Engineering Thermodynamics Previous Mid-1 & Semester Examination Papers

Course Code:	R23-MEC-1204	**Academic Scheme:**	I B.Tech II Semester / II B.Tech I Semester
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Section A: Zeroth & First Law Applications

Microscopic vs Macroscopic approach, Quasi-static process, Heat and work transfer, First Law applied to non-flow and steady-flow processes (SFEE), PMM-1.

Section B: Second Law of Thermodynamics & Entropy

Kelvin-Planck and Clausius statements, Carnot cycle, Reversible and irreversible processes, Clausius inequality, Entropy generation and principle of increase of entropy.

Section C: Properties of Pure Substances & Steam Tables

p-v-T surfaces, Mollier diagram, Critical state and dryness fraction determination, Steam calorimeters, Throttling process.

Section D: Thermodynamic Gas Power Cycles

Air standard Otto, Diesel, and Dual combustion cycles, Mean effective pressure, Thermal efficiency comparison for given compression ratio and peak pressure.

Section E: Vapor Power & Refrigeration Fundamentals

Simple Rankine cycle, Reheat and regenerative modifications, Vapor compression refrigeration cycle (VCRC), COP analysis.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Exam Duration	3 Hours (Total Marks: 70)
Part A (Compulsory)	5 Questions x 4 Marks = 20 Marks
Part B (Unit Choice)	5 Questions x 10 Marks = 50 Marks (Internal choice from each unit)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM